b) **Mechanical Branch**- (Core 70 Marks)

Strength of Materials: Simple stresses and strains Hooke's law, elastic constants, stress strain curve of mild steel bars of uniform strength, compound bars, temperature stresses, stresses on oblique planes – principal stresses and strains, Mohr's stress circle, shear force and bending moment diagrams for beams, bending and shear stresses in beams, deflections of beams, columns and struts, strain energy, torsion of circular shafts and springs.

Fluid Mechanics and Machinery: Basic fluid properties, fluid static – pressure measurements, buoyancy and flotation, fluid kinematics, fluid dynamics – Euler's Bernoulli's and Impulse momentum equations, laminar and turbulent flows, flow through pipes and losses in pipes, bends, boundary layer theory, compressible fluid flow, impact of jets, Hydraulic turbines and pumps, Ram, Accumulator and intensifier.

Material science and metallurgy: Structure and properties of engineering materials, bonding in solids, imperfections in crystals and metals, structure of alloys, manufacture of iron and steel, heat treatment, alloy steels, principles of powder metallurgy.

Theory of Machines: Displacement, velocity and acceleration analysis of plane mechanisms, dynamic analysis of slider – crank mechanism, gear trains, flywheels.

Vibrations: Free and forced vibrations, effect of damping, resonance, vibration isolation, critical speeds of shafts.

Design of Machine elements: Design for static and dynamic loading failure theories, fatigue strength, S-N diagram, design of joints, shafts, bearings, gears, brakes, clutches, screws, springs, cranks, piston, gyroscopes, balancing and governors.

Heat Transfer: Various modes of heat transfer, fins, heat exchangers, LMTD & NTU methods, unsteady state heat conduction, dimensionless parameters, free and forced convective heat transfer, thermal boundary layer, heat transfer in flow-over flat plates and through pipes, effect of turbulence, radiative heat transfer, shape factors, network analysis, condensation and boiling.

Thermodynamics: Zeroth, first and second laws of thermodynamics, thermodynamics systems and processes, Carnot cycle, Air-standard cycles, irreversibility and availability, properties of pure substances, psychometry, Refrigeration and Air conditioning, working principles and their applications.

Applied Thermodynamics: Classification of compressors and its working principles, Classification of I.C. Engines and its working principles, performances, Design considerations of combustion chambers for C.I. & S.L Engines, knocking, rating of fuels, lubrications, Ignition systems.

Turbo Machines: Working principles of gas turbines, steam turbines, Rankine's cycle. Modified Rankine's cycle, jet propulsion and nozzles.

Metal cutting and machine tools: Mechanics of machining, single and multi-point cutting tools, tool geometry, tool life wear, cutting force analysis, micro finishing machines – EDM, ECM and USM, NC machines, jigs and fixtures. Standards of measurements, limits, fits, tolerances, linear and angular measurements, comparators, lathes, drilling, shaping, planning, milling, gear cutting. Broaching and grinding machines

Foundry, Welding and Forging: Design of patterns moulds and cores, solidification, design consideration of runner, riser and gate. Physics of welding, types of welding and their principles, brazing, soldering, adhesive bonding, Fundamental of hot and cold working processes, forging, rolling, extrusion, drawing, shearing and bending.

Production and operation management: Plant layout, material handling, production planning and control, materials, management and work studies, inspections, quality control, cost analysis, operation research, basic concepts of CAD/CAM, inventory control.

e) **Common Syllabus for all exams**"

(i) Reasoning and General Intelligence: Questions of both verbal and non-verbal type. These will include questions on Semantic Analogy, Symbolic operations, Symbolic/Number Analogy, Trends, Figural Analogy, Space Orientation, Semantic Classification, Venn Diagrams, Symbolic/ Number Classification, Drawing inferences, Figural Classification, Punched hole/ pattern-folding & unfolding, Semantic Series, Figural Pattern-folding and completion, Number Series, Embedded figures, Figural Series, Critical Thinking, Problem Solving, Emotional Intelligence, Word Building, Social Intelligence, Coding and de- coding, Numerical operations, Other sub- topics, if any.

(ii) General Awareness: Questions are designed to test the candidates" general awareness of the environment around them and its application to society. Questions are also designed to test knowledge of current events and of such matters of everyday observation and experience in their scientific aspect as may be expected of an educated person. The test will also include questions relating to India and its neighboring countries especially pertaining to History, Culture, Geography, Economic Scene, General policy and scientific research.

(iii) Quantitative Aptitude: The questions will be designed to test the ability of appropriate use of numbers and number sense of the candidate. The scope of the test will be the computation of whole numbers, decimals and fractions and relationships between numbers. It will test sense of order among numbers, ability to translate from one name to another, sense or order of magnitude, estimation or prediction of the outcome of computation, selection of an appropriate operation for the solution of real life problems and knowledge of alternative computation procedures to find answers. The questions would also be based on arithmetical concepts and relationship between numbers and not on complicated arithmetical computation (The standard of the questions will be of 10+2 level).

(iv) English Language And Comprehension: Vocabulary, grammar, sentence structure, synonyms, antonyms and their correct usage; Spot the Error, Fill in the Blanks, Synonyms/ Homonyms, Antonyms, Spellings/ Detecting mis-spelt words, Idioms & Phrases, One word substitution, Improvement of Sentences, Active/ Passive Voice of Verbs, Conversion into Direct/ Indirect narration, Shuffling of Sentence parts, Shuffling of Sentences in a passage, Cloze Passage, Comprehension Passage. To test comprehension, two or more paragraphs will be given and questions based on those will be asked. At least one paragraph should be a simple one based on a book or a story and the other paragraph should be based on current affairs editorial or a report.

(v) Computer Knowledge/Proficiency: Computer Basics: Organization of a computer, Central Processing Unit (CPU), input/ output devices, computer memory, memory organization, back- up devices, PORTs, Windows Explorer, Keyboard shortcuts. Software: Windows Operating system including basics of Microsoft Office like MS word, MS Excel and Power Point etc.. Working with Internet and e-mails: Web Browsing & Searching, Downloading & Uploading, Managing an E-mail Account, e-Banking. Basics of networking and cyber security: Networking devices and protocols, Network and information security threats (like hacking, virus, worms, Trojan etc.) and preventive measures.